Power BI is a Business Intelligence Tool for anyone who needs to turn data into information. It offers data modeling, data visualization & data sharing.

You get to create your own reports without waiting for IT to set these up. It's designed for non-techies although techies love it too.
You get to be in charge and define your data preparation process, data model and design reports.

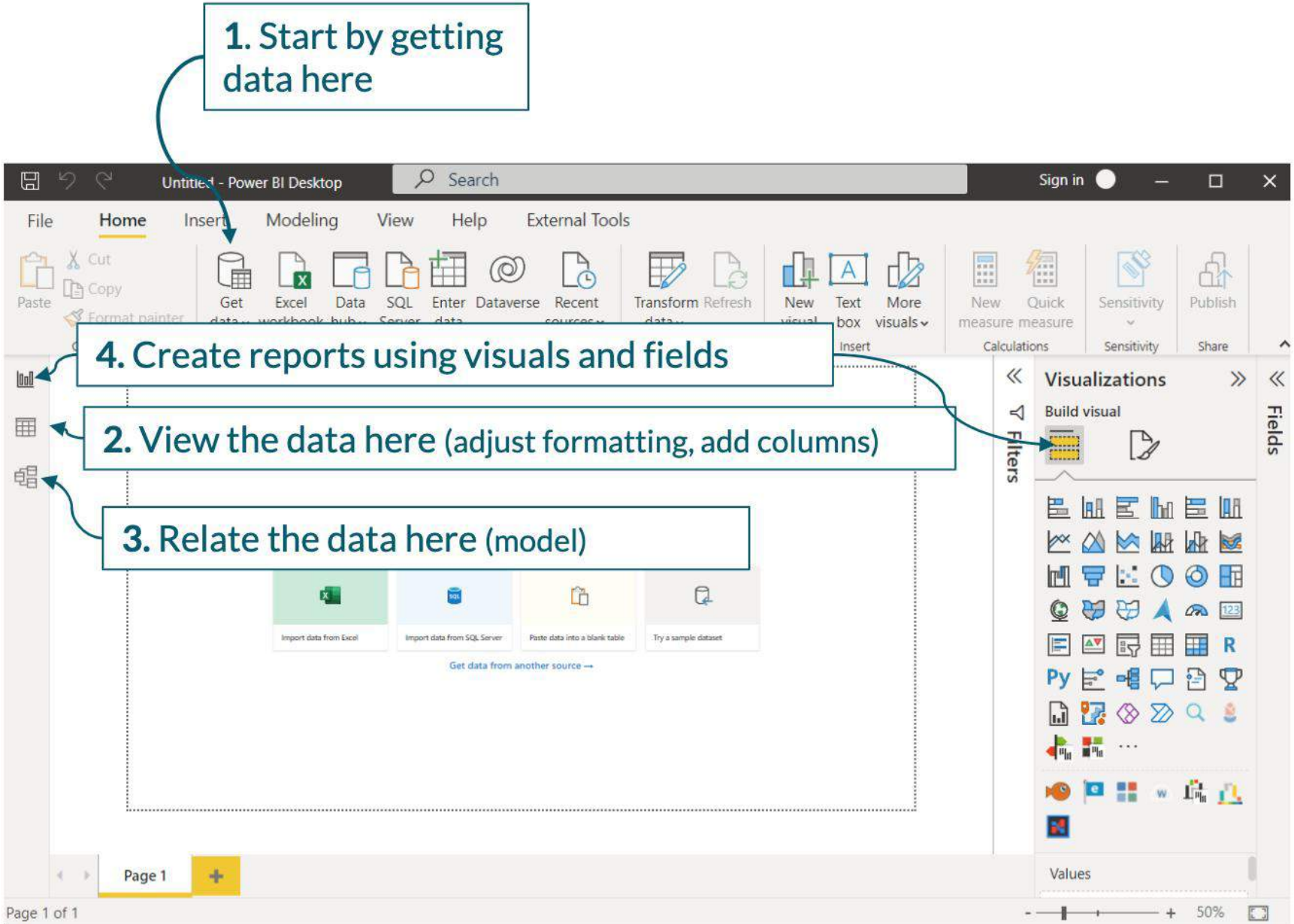
POWER BI Desktop: Free Development tool of Power BI Reports. There, you connect to your data, clean it, model it, add calculations and visualizations. You can save your work in a .pbix file, and you can publish it to the cloud or to an on-premises server.

POWER BI Service: Cloud method of sharing Power BI reports and dashboards. Available for a monthly subscription fee. Limited version also available for free. Requires a "work" email address.

POWER BI Report Server: On-premises Report Server that enables sharing of Power BI reports for those that don't want cloud deployment. Available as subscription as part of Power BI Premium.

POWER BI Mobile: Free mobile app available for iOS, Android and Windows. Allows you to connect and interact with your cloud and on-premises data on the go.

Power BI Desktop: High Level Overview



Every Power BI project begins by connecting to your source data (1).

You can then view the data in the data tab (2) and adjust formatting, update column categories (for example whether a column is a URL or a geographical location) and add new columns.

You can relate the data from different sources and tables with one another in the Model tab (3). This is where you define the relationships between the data (if you're using multiple tables).

Finally, you create your reports in the report tab (4).

Quick Tour of Power BI Desktop

Most Common Features For Every Menu Option

File

Options and settings

control some important functions, such as autodetection of relationships and data types, security & language settings, and preview features, among others.

Home

Format painter – easily copy and paste formatting between visuals

Get data– connect to a source

Enter data – manually enter data, create a new table

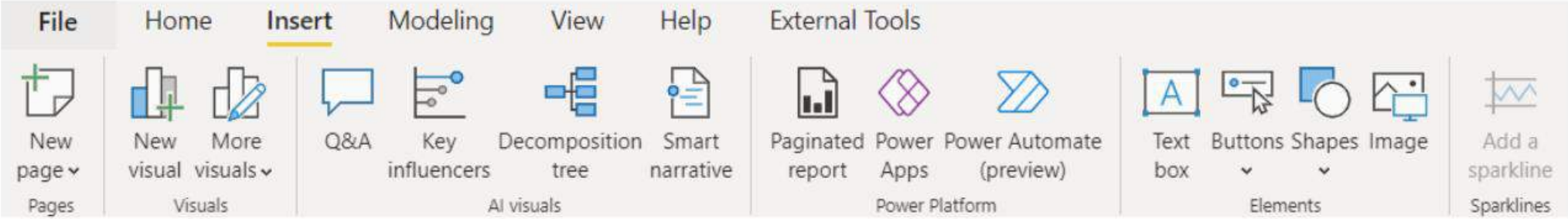
Transform data– open Power Query editor

(+ Data source settings – change the source path)

Refresh – get the latest data from the source

Publish – send your report to Power BI service, where it can be securely stored and shared with others

Insert

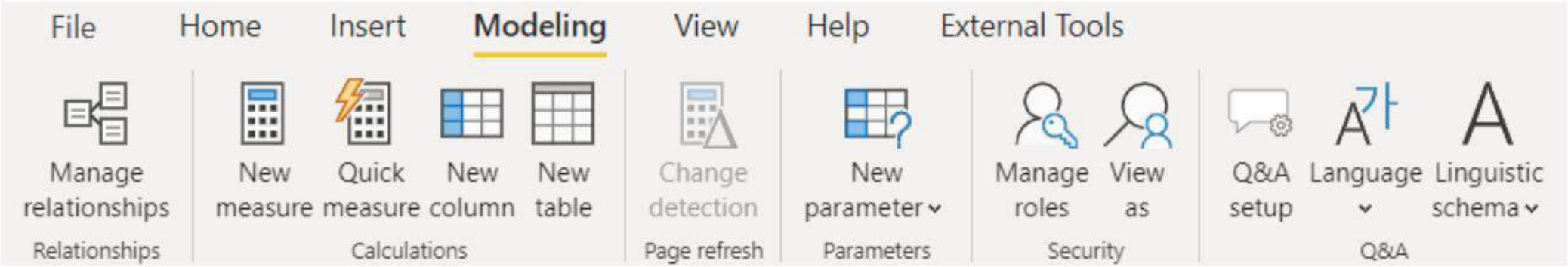


AI visuals

Elements - Text box, Buttons, Shapes, Image

Quick Tour of Power BI Desktop

Modelling

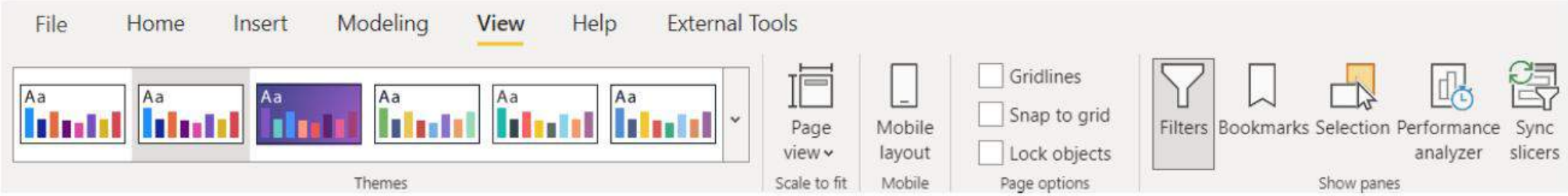


Calculations -New/Quick measure, New column, New table

New parameters (Fields)

Security - Manage roles & View as

View



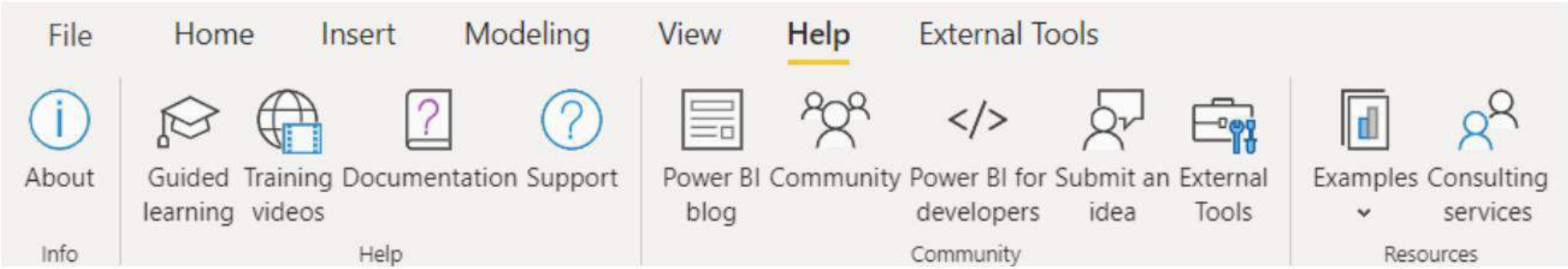
Themes – change the design of your report with 1 click

Mobile layout – optimize your report design for a mobile screen

Show panes - Bookmarks, Selection, Performance analyzer

Sync slicers – synchronize slicer selection between pages

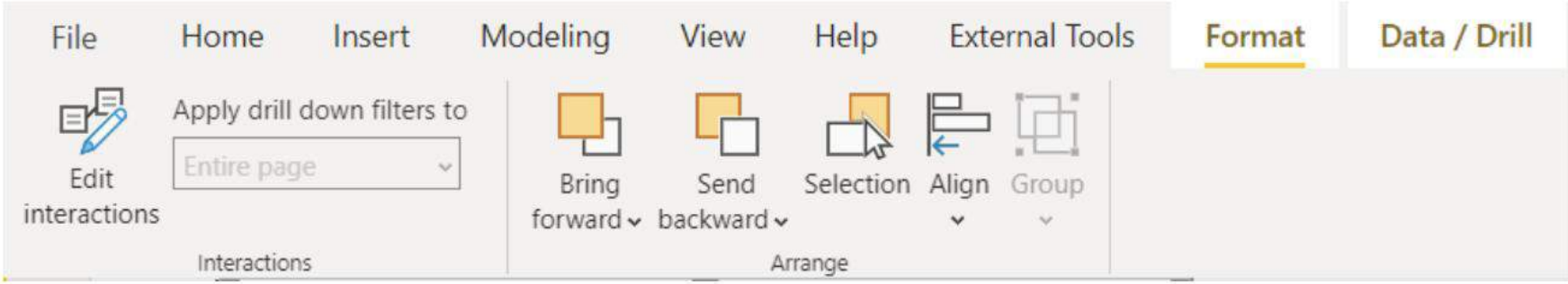
Help – array of useful resources



Quick Tour of Power BI Desktop

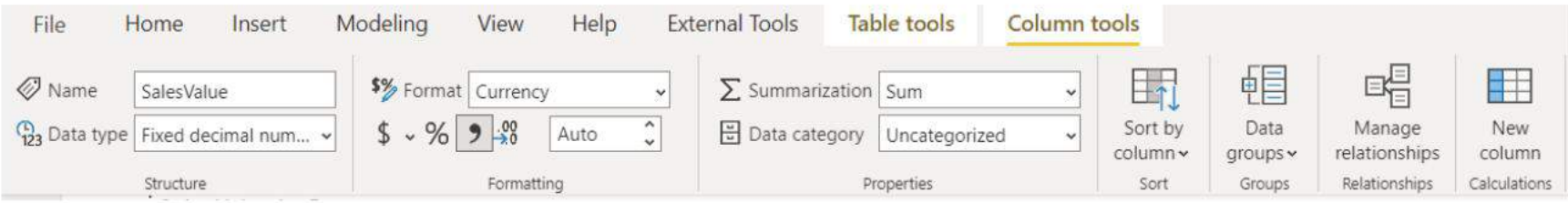
Contextual Menus (dependent on selection)

Format (with visual selected)



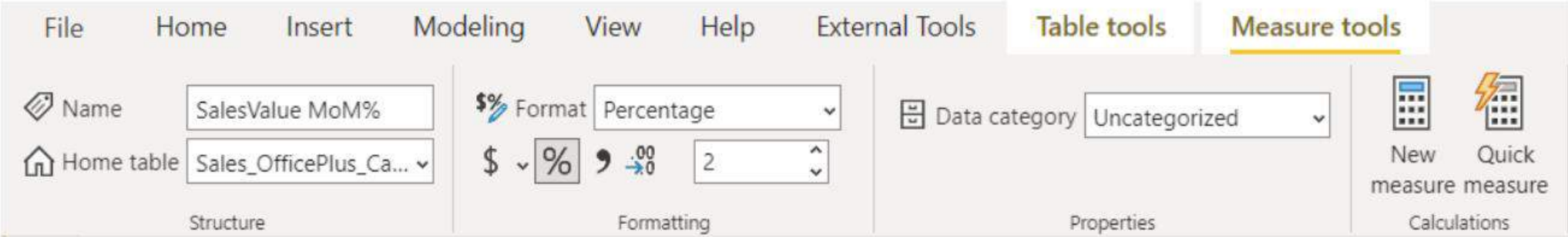
Column tools (with column selected)

adjust column formatting assign a data category, change the default summarization, define custom sort order



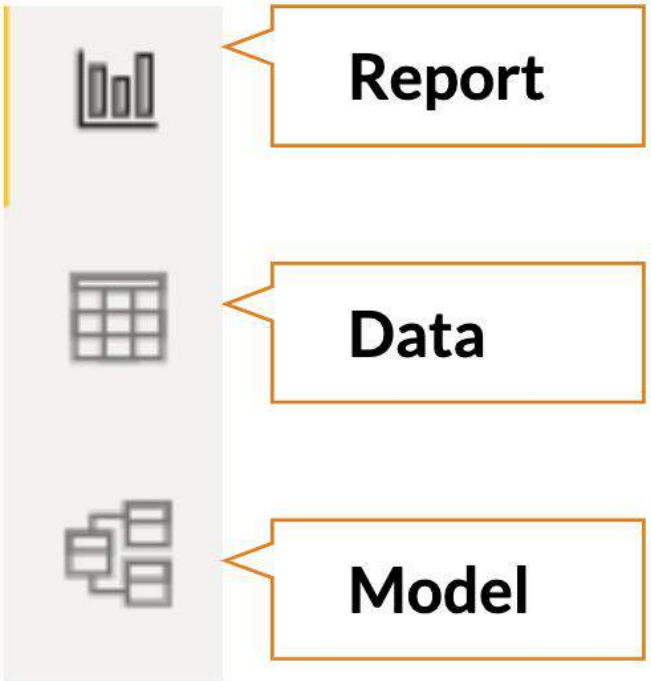
Measure tools (with measure selected)

adjust measure formatting, move to another table, or rename it



Quick Tour of Power BI Desktop

You will find yourself operating between the 3 principal views:

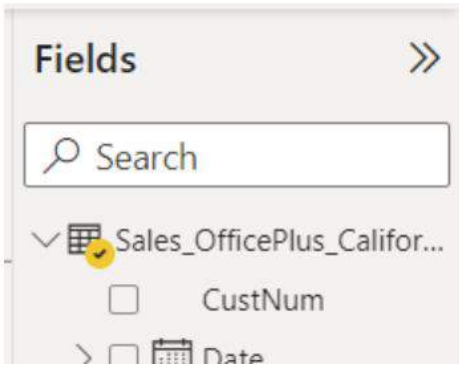


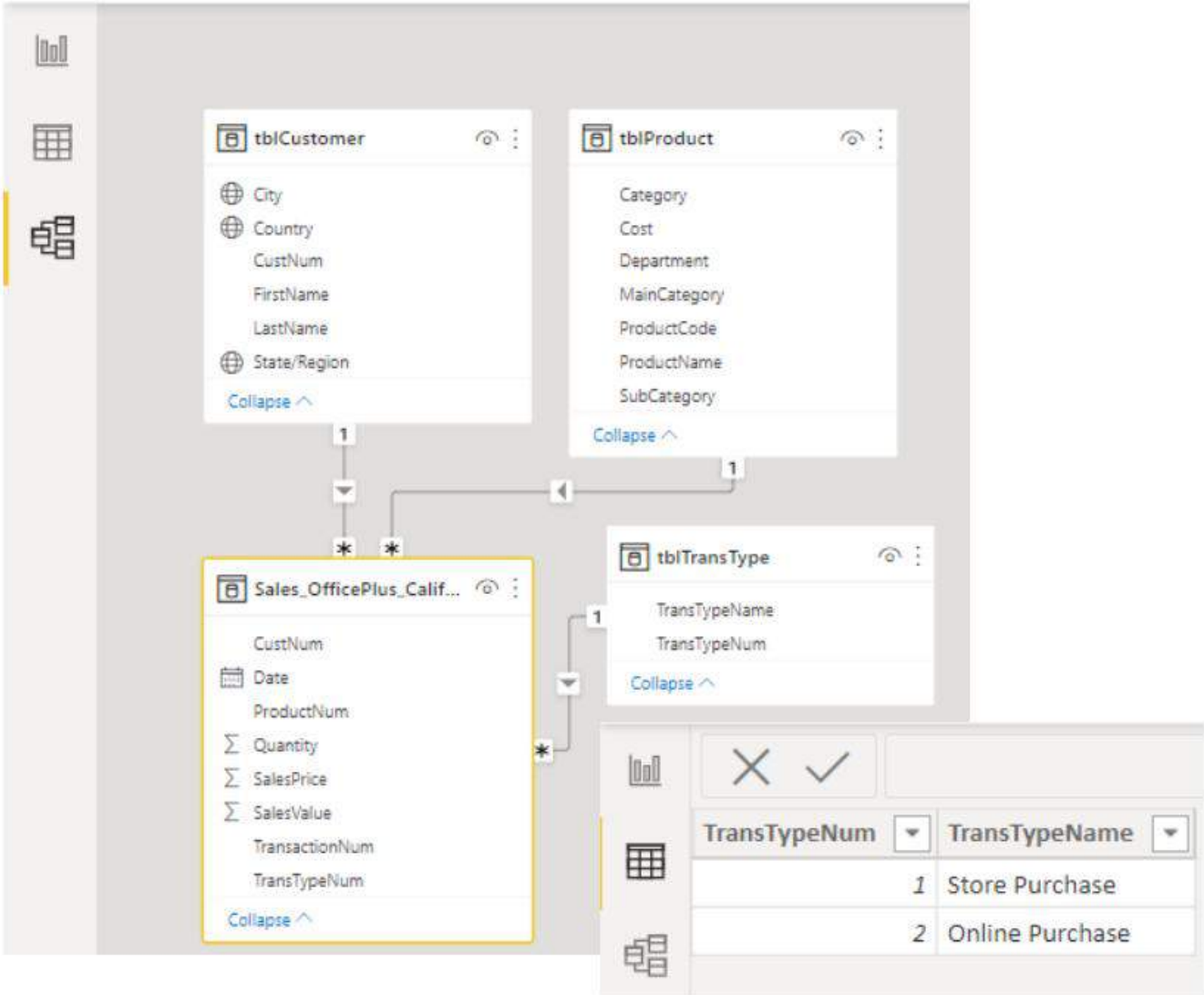
This is where you create visuals and build the overall report.

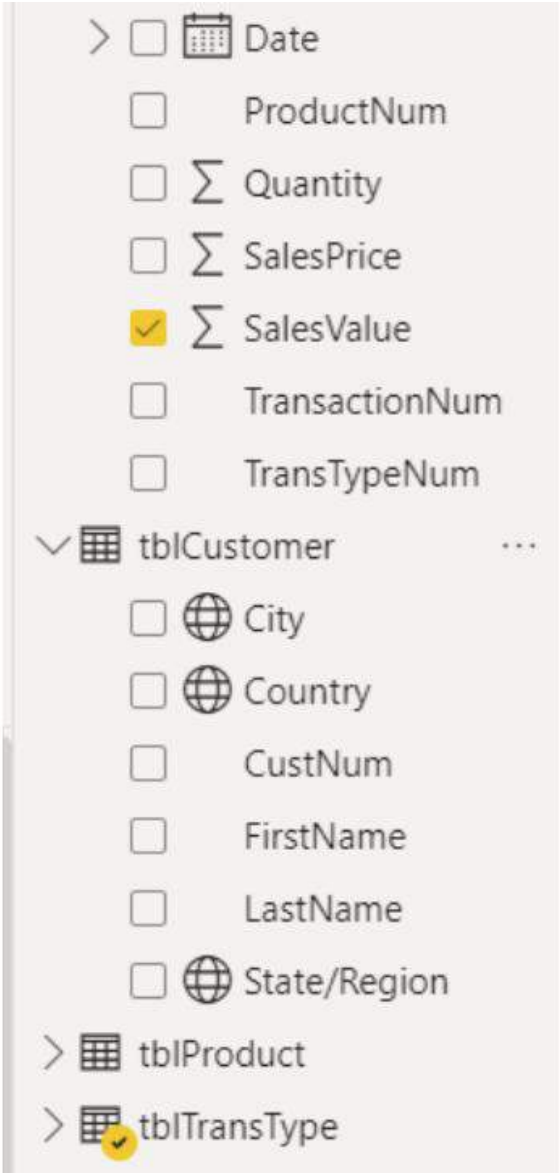
Here you can preview your data, assign the formats, categories and summarizations to columns/fields.

In this view you add and manage relationships between tables, shaping them into a data model.

In all views you get the **Fields** pane, where you can edit fields' properties by right-clicking, and (from Report view) add fields to visualizations.




tblCustomer
City
Country
CustNum
FirstName
LastName
State/Region
Collapse
tblProduct
Category
Cost
Department
MainCategory
ProductCode
ProductName
SubCategory
Collapse
Sales_OfficePlus_Calif...
CustNum
Date
ProductNum
Quantity
SalesPrice
SalesValue
TransactionNum
TransTypeNum
Collapse
tblTransType
TransTypeName
TransTypeNum
Collapse
1
1
1
*
*
*
TransTypeNum
TransTypeName
1
Store Purchase
2
Online Purchase


Date
ProductNum
Quantity
SalesPrice
SalesValue
TransactionNum
TransTypeNum
tblCustomer
City
Country
CustNum
FirstName
LastName
State/Region
tblProduct
tblTransType

Proper Data Set

To conduct any analysis, you need clean data, organized in a proper data set. What does it mean?

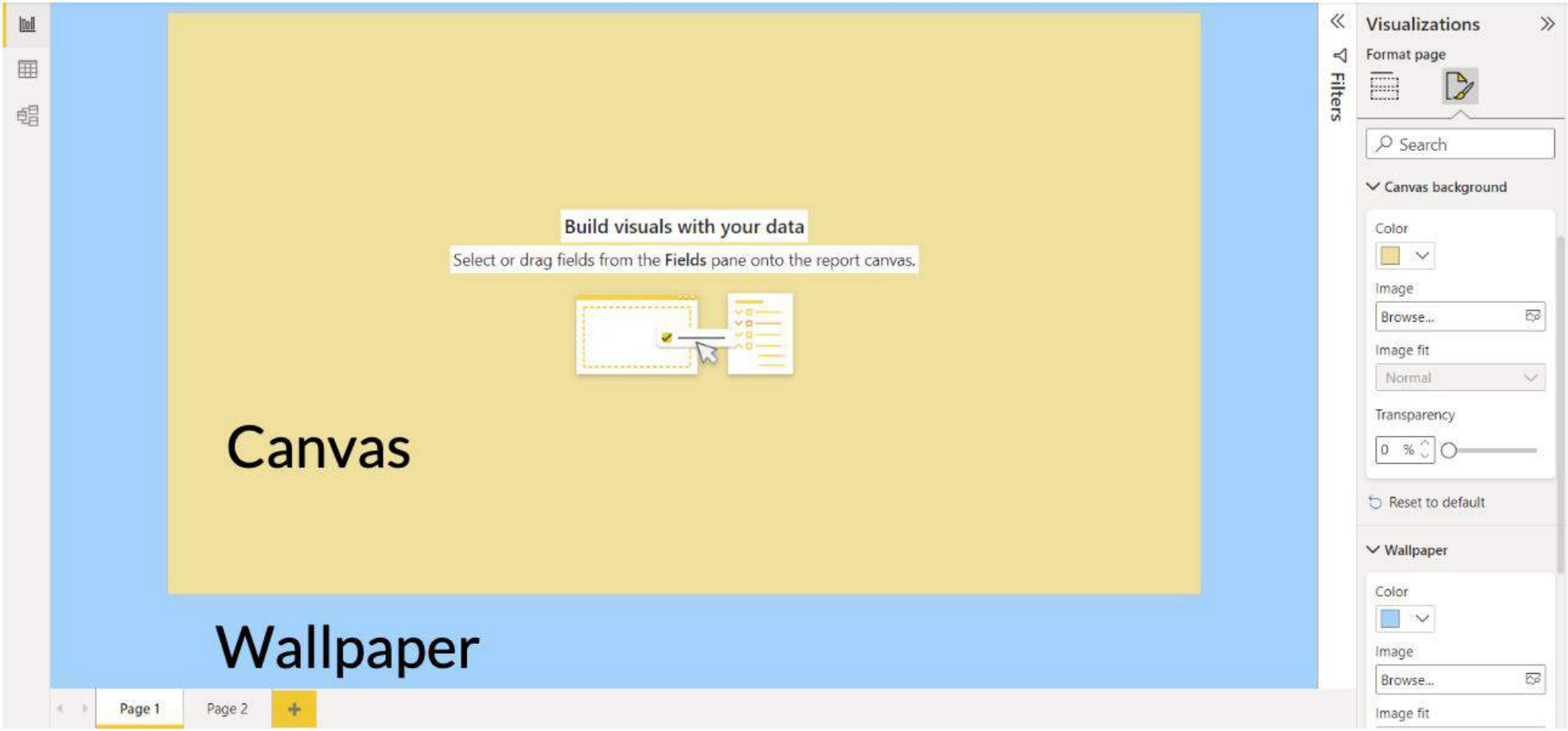
1. **Each column has a single purpose** (you don't mix customers and countries in the same column)

2. **There are no empty rows & columns**

3. **Each column has one single data type** (so either date, number, text, but not a mixture of those in a single column)

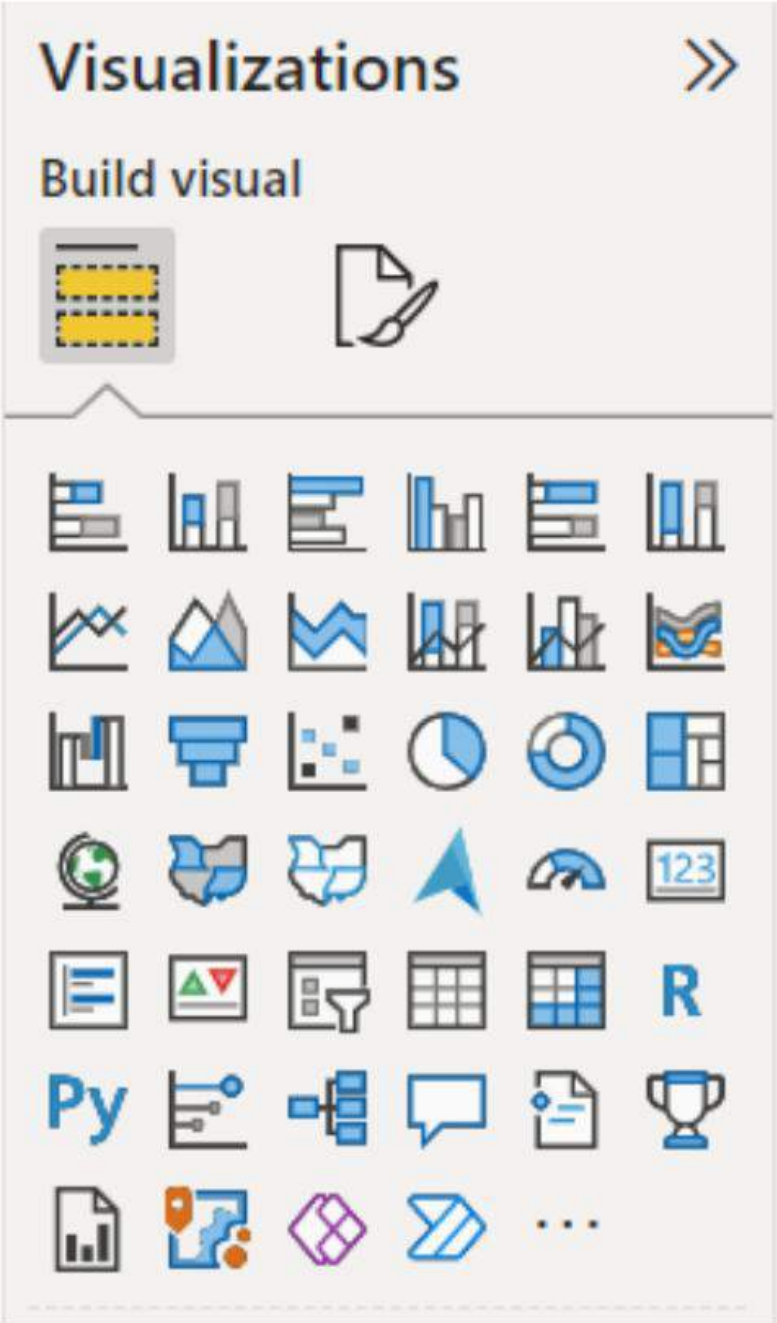
If your source data is not structured this way, Power Query is the perfect tool to clean it up.

Visuals


Build visuals with your data
Select or drag fields from the Fields pane onto the report canvas.
Canvas
Wallpaper
Page 1
Page 2
Filters
Visualizations
Format page
Search
Canvas background
Color
Image
Browse...
Image fit
Normal
Transparency
0 %
Reset to default
Wallpaper
Color
Image
Browse...
Image fit

You add visuals to the **canvas** by inserting them from the Visualizations pane on the right and populate them with fields from the Fields pane. Visualizations pane also includes the **formatting** options.

If you start building visuals by inserting a field, the default type depends on the field you use. **Clustered column chart** if you start with a numeric value, **Table** if you start with category, **Map** if you start with geography field.

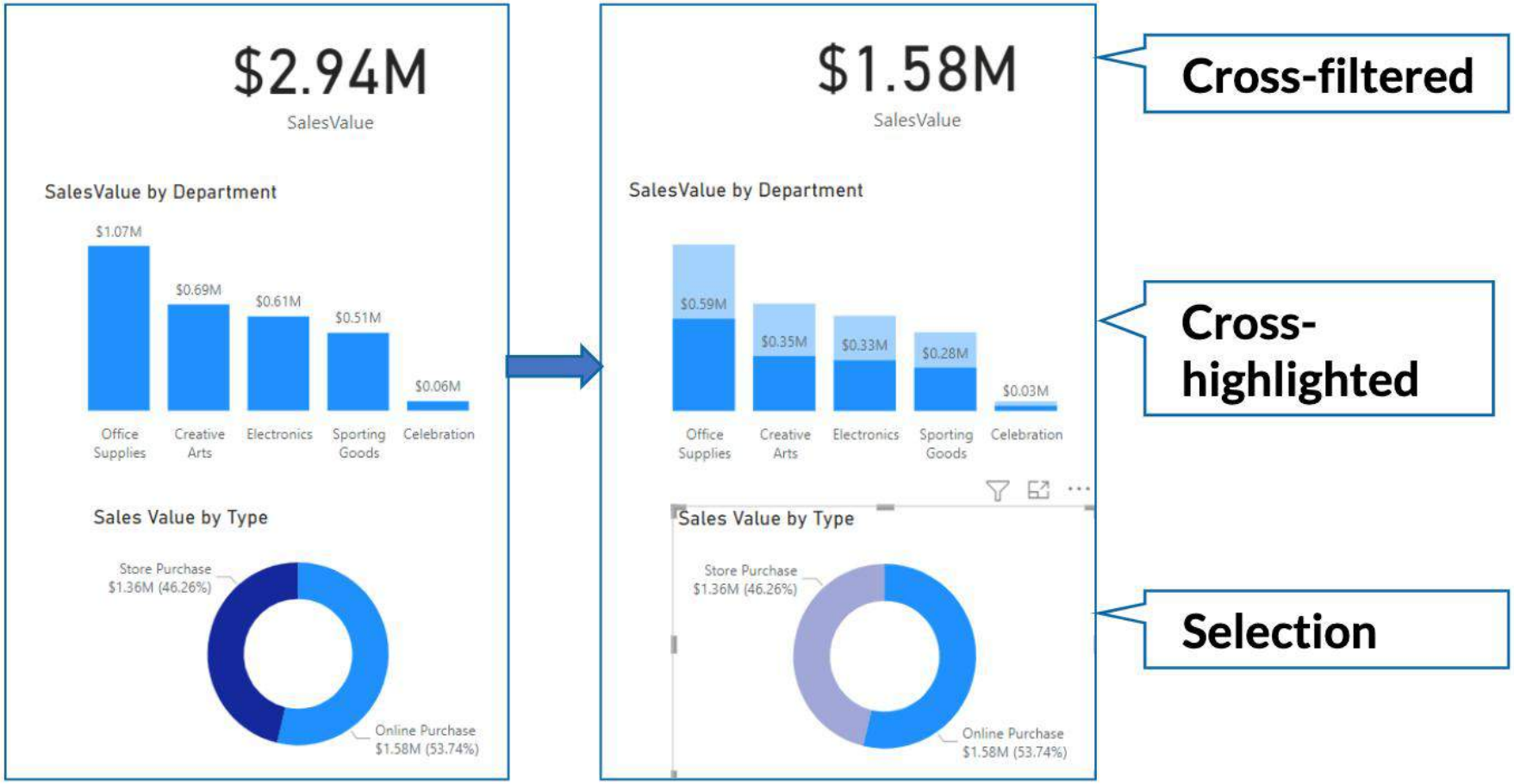


Visuals / Filtering

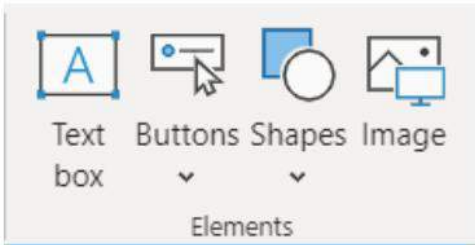
Visuals interact with each other – you can filter or cross-highlight a visual by making a selection on another visual.

Cross-filtering removes data that doesn't apply.

Cross-highlighting retains the original data points but dims the portion that doesn't apply.



In addition to the various charts and slicers, from the Insert tab on the ribbon you can add other visual elements, such as text boxes, shapes and images.



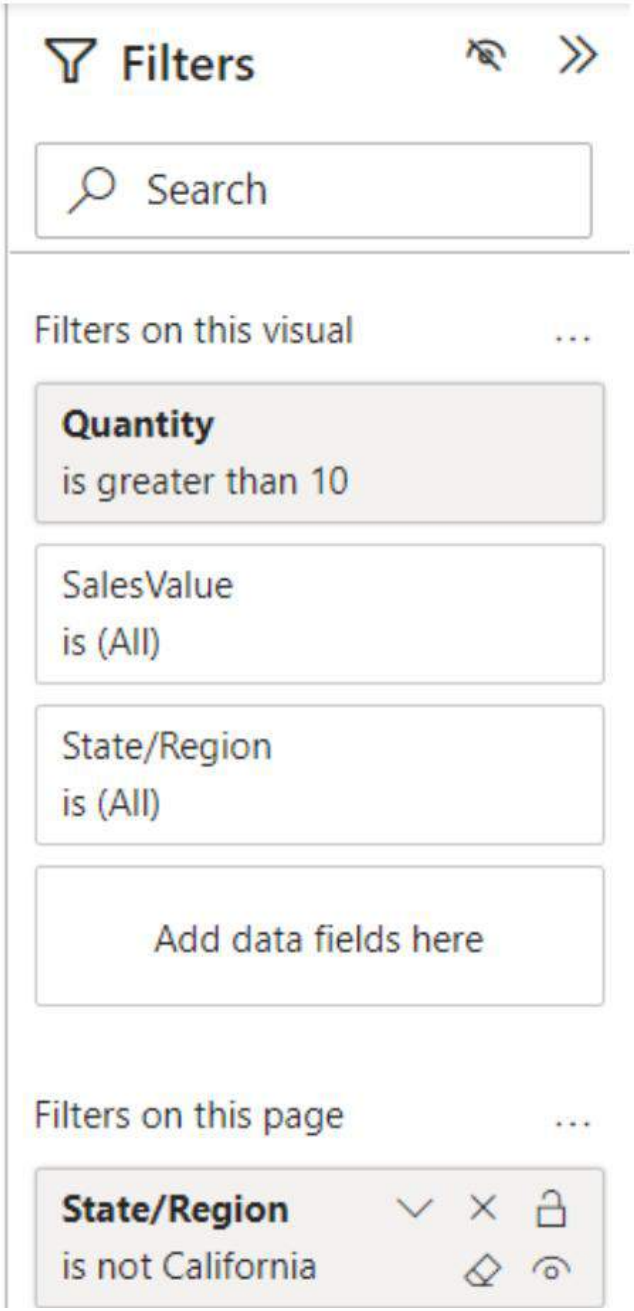
Filters Pane

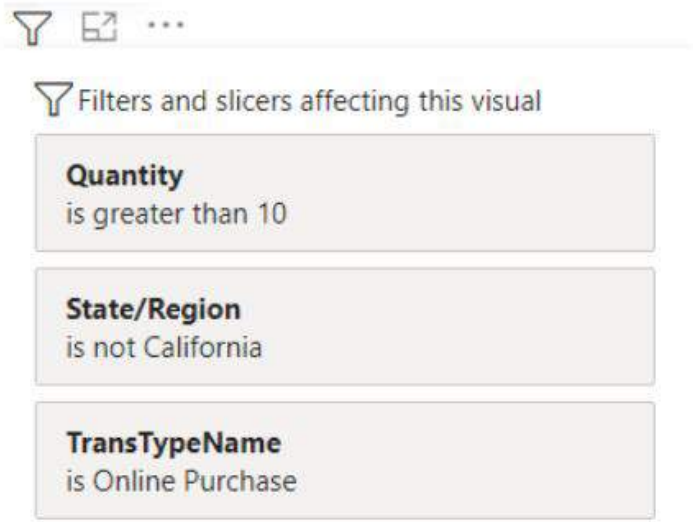
Using the filter pane to the right of the canvas, you can apply filters on:

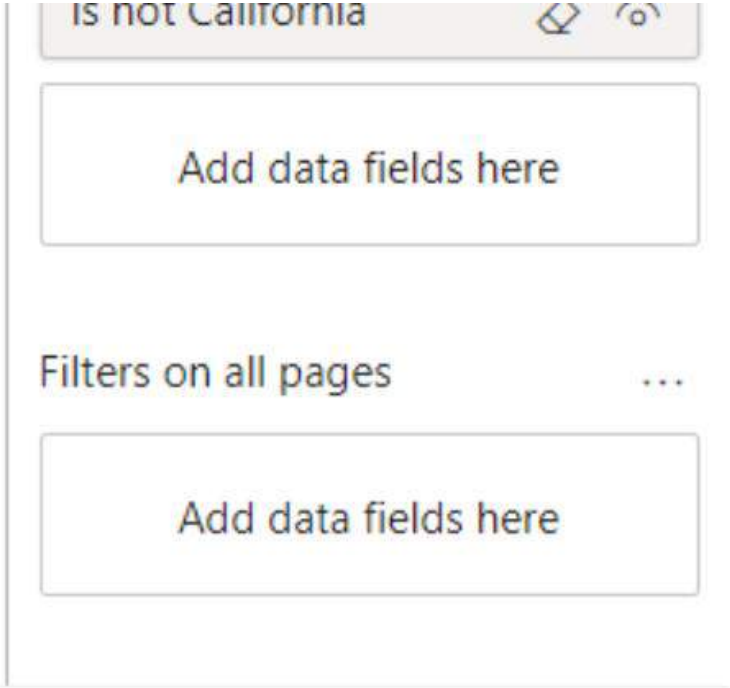
- Selected visual (visual level filter)
- All visuals on the current page (page level filter)
- All visuals on all pages of the report (if more than 1) (report level filter).

There are several filter types: Basic, Advanced, Top N, Relative date and Relative time (for date/time fields only).

You can see all filters impacting a visual by hovering over the funnel icon next to it.







You can collapse the filter pane,
you can hide it (it won't be visible to report consumers when the report is published).
You can also hide a specific filter or lock it
(report consumers won't be able to change it).
To clear filter, use the eraser icon.

Slicers

Slicers are visuals that allow you to filter other visuals.

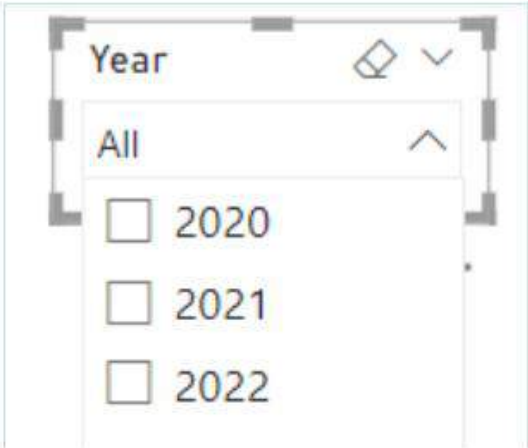
They are more interactive than the Filter pane, as they are part of the report.

You have several types to choose from

List Slicer (multiselect vs single select)

Button (List Slicer with Horizontal orientation)

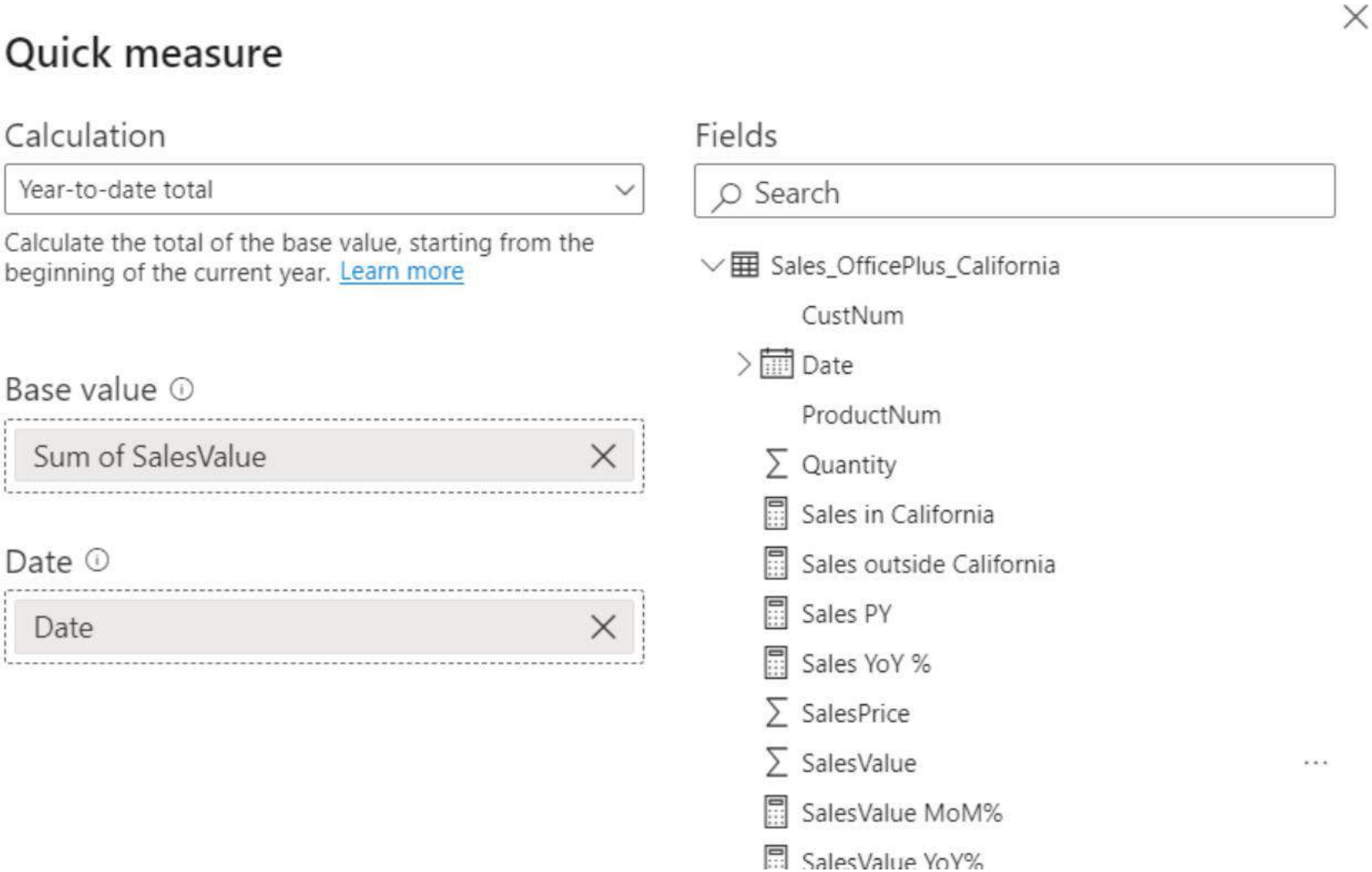
Dropdown



Between / Less than or equal to / Greater than or equal to

Quick Measures

A quick measure is a type of explicit measure that is created using a template. You can create calculations without prior knowledge of the DAX formula language.



TransactionNum

TransTypeNum

> tblCustomer

> tblProduct

> tblTransType

Post an idea

OK

Cancel

Glossary

DATA CATEGORY – The **description** of the data. Helps to contextualize the data, and translate it to visuals, for example geographic classification, or Web URL.

DATA FORMAT – The **display** of data. This is the "makeup" you apply to the data to define how it should be "seen". Remember, data type and data format are NOT the same thing.

DATA MODEL – a visual representation of data, how it connects to each other and flows. In Power BI, data modeling means creating relationships between tables – usually a fact (data) table and dimension (or lookup) tables.

DATA TYPE – The **type** of data. This decision also impacts the precision of the values. For example, currency data type keeps 4 decimal places in memory, whereas whole number keeps 0 decimal places. You assign it to a column either in Power Query Editor or in Power BI. More data types are available in Power Query than in Power BI Data view.

EXPLICIT MEASURE – a calculation created by you, using DAX, Data Analysis Expressions. Includes QUICK MEASURES, which are templates that generate the DAX code using fields supplied by you.

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IMPLICIT MEASURE – a calculation that is applied automatically when numeric fields are added to a visual, using one of the available aggregations (SUMMARIZATION).

POWER QUERY – data transformation and preparation engine used in Power BI (and Excel, among others). Includes a graphical interface for getting data from sources and a Power Query Editor for applying transformations.

RELATIONSHIP – the connection between tables. Has **cardinality** (e.g. one to many), and **cross filter direction** (single or both ways).

SUMMARIZATION – how numeric fields are aggregated: Sum, Average, Minimum, Maximum, Count or Distinct Count.